16-bit Image-Process `Img16Prc` (for Windows, Ubuntu) Deployment archive for Evaluation .

2026-Nov by S.Bochkarev

1. The review

1.1. The functionality:

This is simple portable approach to be an equivalence of sophisticated image-presentation-enhancements in End-User-Applications (such as Radiography Consoles).

The input:
- a "`*.raw`" (16-bit) original-image (`*.DCM`, `*.tif`, etc for ImageJ),
- chosen "processing mode",
- chosen "adjustments".

The output:
- processed image of the same dimension.

1.2. The files:

The binaries
`Img16Prc.exe`, `Img16Prc.dll` - Windows;
`Img16Prc`, `Img16Prc.so` - Linux.

Initialization profiles (configuration files):
`Img16Prc??.ini` (where "`??`" is ~`sprintf("%2d",process_mode)`)

Sample source images:
`???0` - calibrated_ nonprocessed.
`??50` - residuals of anti-scatter grid.
`??60` - presense of anti-scatter grid.

Sample <to compare against> processed images :
`???1` - known "state-of-art" process for `???0`.
`???2` - known "High Dynamic Range filter" of `???1`.
`???3` - known "antiscatter grid filter" of `???2`.
~`???std` - known "standard process" in End-Use-App.

Sample <resulting> processed images (might <occacsioally> exist in a deployment archive):
`????_MM` or `????_MM_TBCS` process mode MM (with adjusments TBCS).

ImageJ-macro:
`Img16Prc.ijm`

Processing scripts (here XYZ=IFR - infrared, =MRI- magnetic resonance, etc):
_exeXYZ.bat|sh - process available samples *.raw (from application-domain "XYZ") with most default TBCS "0888" or "1888". For each original *.raw generate single processed image *_MM_0888.raw (or *_MM_1888.raw respectively).
_exeXYZall.bat|sh - generate variety of processed images.

Service scripts:
_clean.bat|sh - delete newly generated processed images (by the wildcards mentioned just above).

Service *.raw data-series viewer (one could also import a *.raw into ImageJ):
Img16DatSer.exe.

2. Deployment

2.1. The materials

Out_Img16Prc.zip - deployment_archive;
_unzip.bat - deployment_script Windows;
_unzip.sh - deployment_script Linux.
The deployment_script decompresses deployment_archive to target deployment_directory and then deletes irrelevant (for specific OS) files there.

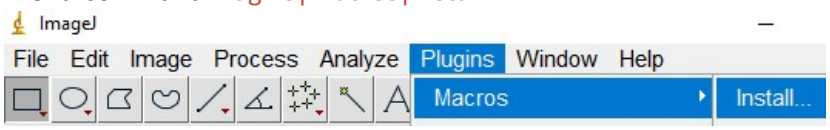
2.2. The deployment's flow

- Set target deployment_directory (in deployment_script edit the line "*set TGT=e:\Img16Prc*" or "*TGT=~/Img16Prc*"). The deployment_directory shall not exist yet (for unambiguity).
- Execute the deployment_script.

Suggested deployment_directories:
c:\Img16Prc\, d:\Img16Prc\, e:\Img16Prc\ - Windows,
~/Img16Prc\, /mnt/D/Img16Prc\ - Linux.
The exact name of deployment_directory is important for ImageJ-macro (which searches binaries only in few explicit paths). One could alter those paths in the text of Img16Prc.ijm. However the scripts .bat, .sh should work anywhere.

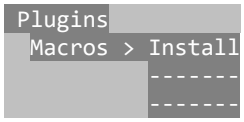
2.3. Integration into ImageJ (ver ~>"1.53t")

Menu-command Plugins|Macros|Install ...

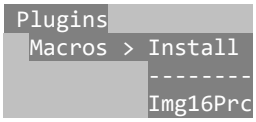


Invoke dialog [ ] "Install Macro"
Look in: d:\Img16Prc* Browse for Img16Prc.ijm (which

```
Img16Prc.dll                    is a file in deployment directory
Img16Prc.exe                    ~"d:\Img16Prc\Img16Prc.ijm").
Img16Prc.ljm
```



New item becomes visible in the submenu [Plugins][Macros]



The macro needs an "opened 16-bit grayscale image" to run. At 1st, it displays a dialog specifying "process mode" and "adjustments".

~Process modes:

```
------------------------------ This sketch depicts "processing modes"
| 20 Lims                    | in a list-box. The modes match to the
| 22 Skull lateral           | files "Img16Prc??.ini" in the deployment-
------------------------------ directory.
```

~Adjustments:

```
-- Tag ---------------------- This sketch depicts check-box, which filters
| [x] anti-scatter grid      | anti-scatter grid. Specific implementation might
------------------------------ have other check-boxes.

-- BCS ---------------------- This sketch depicts 3 sliders of range 1..15.
|                 1   8   15 |
| Brightness       ===I===   |
| Contrast         ===I===   |
| Sharpness        ===I===   |
------------------------------
```

Menu-item `Img16Prcs` in `Plugins|Macros` will persist during current ImageJ-session. Permanent installation of this macro needs copying of `/Img16Prc/*` into ImageJ's installation directory (then code-line `File.getDirectory(getInfo("macro.filepath"))` actually begins to work).

3. Sample work-flow

3.1. via command-line and/or processing scripts

This is suitable to process a group of input `*.raw` with fixed adjustments. Execution act is a command-line like the following:

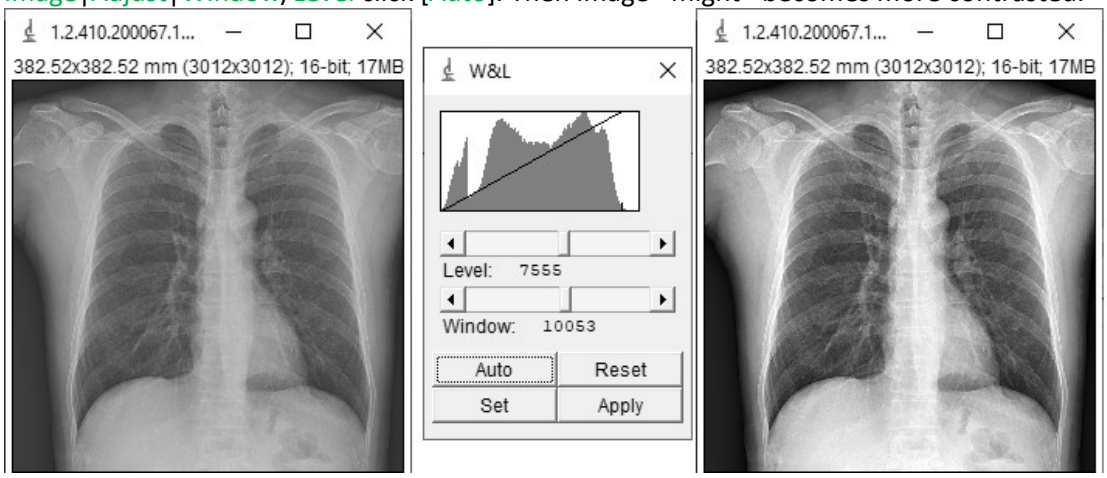
```
%Prog% 3268 3268 Dir/???0.raw 20 1888 04
       |    |    |           |  |    + auxiliary flags
       |    |    |           |  + TBCS
       |    |    |           + process mode
       |    |    + wildcard
       |    + height
       + width
```

3.2. via ImageJ

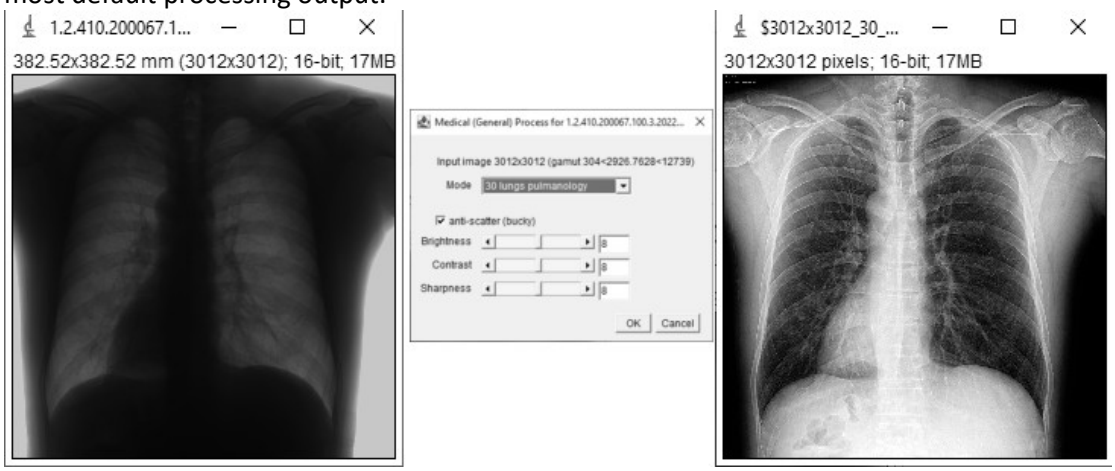
This is suitable to process a single image (to compare against existing sample processed

images, to experiment with the adjustments).

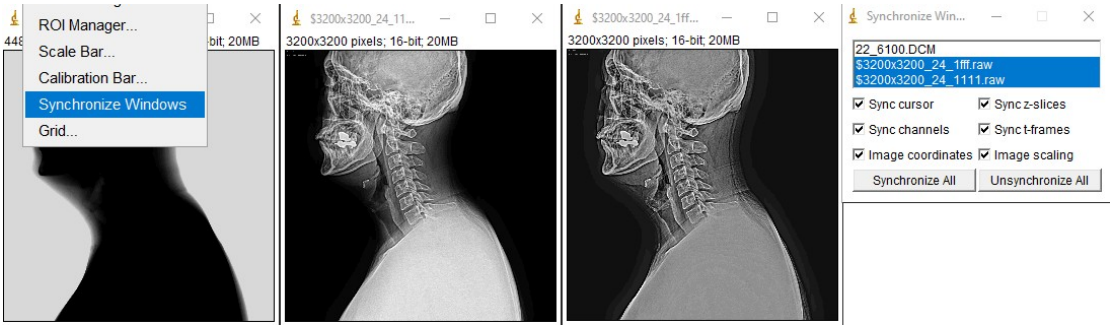
Step 1) Open a sample processed MMM_XYZstd.DCM. In dialog Image|Adjust|Window/Level click [Auto]. Then image <might> becomes more contrasted.



Step 2) Open respective sample calibrated nonprocessed MMM_XYZ.DCM. In Plugins|Macros|Img16Prc select relevant process' Mode and click [Ok] and look at the most default processing output.



Step 3) Comparing the images from previous steps, repeat Plugins|Macros|Img16Prc varying the Adjustments (i.e. the Tag, BCS).



For convenience, one might

- click Analyze|Tools|Synchronize Windows,choose image-windows to compare;
- set the same scale & size of each comparing image-window (by "mouse wheel",

"keyboard-zoom Ctrl+|-");
- compare the image-windows using synchronized "keyboard-zoom" and "mouse-pan by hand-icon".

4. The adjustments

4.1. Encoding "linear run-time adjustments" by triplet BCS

Three 4-bit tags **BCS** simplify presentation enhancement's adjustment replacing <possible functionally equivalent> collection of "similar image processing modes".
B: brightness - most_dark..most_bright output;
C: contrast - mn..mx amplification of ~mid_freqs;
S: sharpness - mn..mx amplification of ~high_freqs.
Tags' values '=0..15' adjust certain parameters, where value '=8' denotes default <the value in initialization profile>, '=**1**' minimizes relevant semantic of the output, value '=**15**=$\mathbf{F}_{hex}$' maximizes one. The meaning of '=**0**' is implementation-specific. The range of numeric parameters' changes by the tags within **1**..**15** should be constrained (perhaps subjectively by developer):
- within a "linearity" of output's effects,
- within rational absolute diapazon of the output.
Hence, **BCS**:
- adjusts ~float weighing-/attenuation-coefficients, ~integer gray-scale-thresholds;
- rather doesn't adjust ~"spatial integers" (e.g. filters' radiuses effect non-linearly).

4.2. Encoding implementation-specific flags via tag T

Tag **T** defines implementation-|datadomain-specific binary flags.

- bit #0 in the most of process modes (~medical X-ray):
=0: disable antiscatter_grid's filter (but if the grid actually exists then it causes high-freq pattern).
=1: enable -"- (but if the grid actually doesn't exist then it causes overblur and wastes time).

- bit #0 in process modes 5,6 (~intraoral):
=0: if resolution is "high" (against certain hard-coded dimension) then process in downsample. Set only for older Android-tablets or high-speed moving acquisition.
=1: accurate (set it normally).

- bit #0 in other process modes (~NDT,MRI):
no effect.

- bit #7: <intentionally undocumented> debug-output.

4.3. Frequently adjusted items in initialization profiles

```
[Settings]
nPrintLabel=-1; print debug label as dark pixels "-16383"
           = 0; don't
           =-1; as bright pixels "16383"
```

[GC] gray-gamut's cut

```
nEnabled=0; sometimes disable (for known image-gamut)
        =1; typically enable
fCutDrkBrt=0.05 0.01; heavy sentinel cut (5% dark, 1% bright pixels)
          =0.005 0.001; less sentinel cut (0.5% dark, 0.1% bright pixels)
```

[CO] collimator detection

```
nEnabled=0; typically disable (to save time)
        =1; enable when peripherals might be collimated
```

[AE] estimate air (i.e. dark background)

```
nEnabled=0; in ~50% situations disable (to save time)
        =1; in ~50% situations enable (when background could be too bright)
```

5. Sample test-flow via ImageJ for single image

5a) By the topic "**2.2**" install macro `Img16Prc.ijm`.

5b) Open 022_6000std.DCM and set automatic gray-scale (topic "**3.2**", **step 1**).

5c) Open 022_6000.DCM and run the macro. Disable check-box "anti-scatter" (as there is anti-scatter grid in this sample). Button [OK] runs `Img16Prc.ijm` with parameters "Mode=22", "TBCS=0888" in command-liine("**3.2**", **step 2**) and produces imageJ's-image-window ~" $3200x3200_22_0888 .raw".

5d) Click 022_6000 to activate again and run the macro with "TBCS=0811".

5e) Click 022_6000 to activate again and run the macro with "TBCS=08FF".

5g) Click 022_6000 to activate again and run the macro with "TBCS=0F88".

022_6000 calibrated source image	022_6000std existing state-of-art	$3200x3200_22_0888 default propsed process
3200x3200_22_0811 mn Contrast,Sharpness	$3200x3200_22_08FF mx Contrast, Sharpness	$3200x3200_22_0F88 mx Brightness



6. Optional large sample data-series with process-variations

6.1. Sample files' names for process-variations

???0 - source calibrated image

???0_NN_TBCS - processing mode NN

- _1833 - low C,S
- _1838 - low C
- _1883 - low S
- _1888 - default TBCS
- _188C - high S
- _18C8 - high C
- _18CC - high C,S

6.2. The purpose

An advantage of this process is the possibility of adjustment of *linear parameters* by End-User via "$T_{ag}B_{rightness}C_{contrast}S_{harpness}$". That can produce few *linear variations* of resulting images (and thus feasibility-tests "**3.1. via BAT-files**", "**3.2. via ImageJ**" are relevant). But when "run-time execution" on applicator's PC is not convenient, then large sample data-series with process-variations could be supplied.

In this situation the rational target of feasibility-test is to "pick up", which specific TBCS is most proper one for specific NN.

7. Desired review<s>/feedback<s>

7.1. The assumptions

Let's assume, that we're comparing presentation enhancement by two data-series (both from the same original images ?o):

data-series ?s	versus	data-series ?o_90
<another> existing enhancement		proposed by this library.

Let's assume, that proposed processing was adjusted for a "best looking" of original image named "5o" (hence the most conscious proposed output of this library is 5o_90).

7.2. The typical targets

typical target **1)** is to determine objective (quantitative|precicise as possible) dictionary of "good versus bad":

- compare adjusted 5o_90.raw vs standard commercial 5s.raw;
- point to 1-3 "worst looking" ROI<s> in 5o_90.raw against 5s.raw.

typical target **2)** is to estimate, what images' features impress subjectively (qualitatively|ambiently perhaps) as "good versus bad":

- compare all processed ?o_90.raw vs standard commercial ?s.raw.
- point "worst looking" image **X**o_90.raw against **Xs**.raw.
- point 1-3 "worst looking" ROI<s> in Xo_90.raw against Xs.raw.

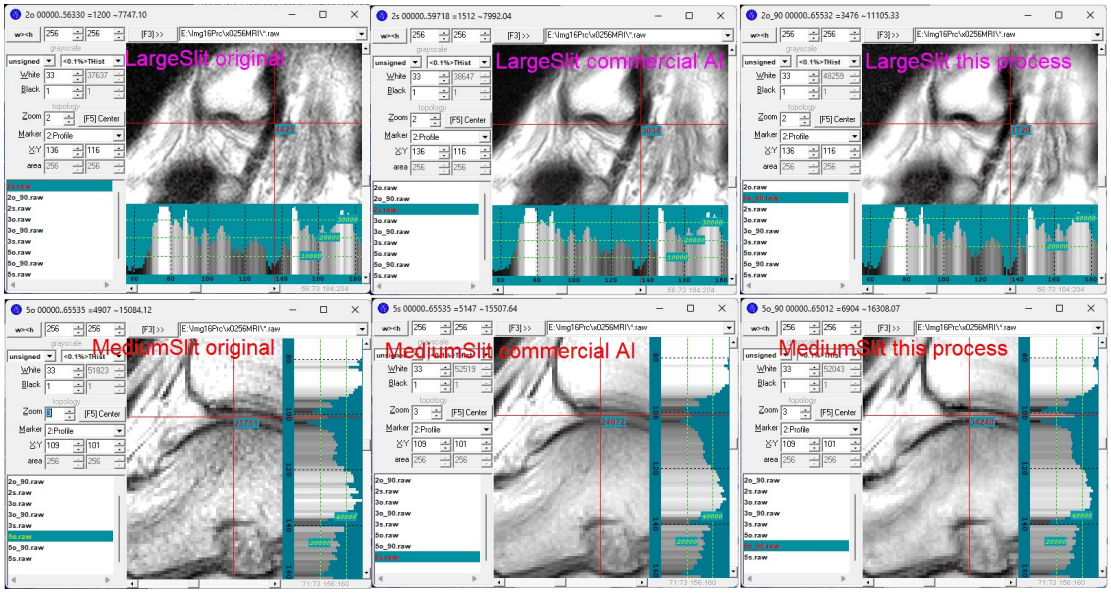
Assume any "convenient" gray-scale viewer here. Lossy JPEG is suitable for screen-shots with meaningful text (PNG,GIF,BMP is OK too). A screen-shot still needs sufficient scale to pass objectivly "overall visual impression" from expert's graphic-device (such as approx gray-window, driver-stretch-mode). A screen-shot centers on the "topic's RoI" and/or contains an <~colored> ~arrow<s>, which point to the "topic's RoI<s>".

7.3. Sample comparison's screen-shots and text

The screen-shots are expecting together with comparison's text in a review/feedback.

~ **Sample comparison Topic 1** *objectively worst-looking-object in this process vs commercial AI process.*

```
The problem is oversharpen LargeSlit's profile. Original noise inside large slits
(inside ~thick joints) should be mitigated (i.e. process should rather blur shape of
LargeSlit's bottom). However the MediumSlit's profile is more sharp (against the
~standard commercial AI process); and this is rather positive feature.
```



~ **Sample comparison Topic 2** *the worst impression from this process vs commercial AI proces.*

The problem is oversharpen backgroud/peripheral_areal. Though this doesn't affect MRI-diagnostic explictly, it worses the comprehension.

